

Kisan-Mitra: Empowering Farmers with AI-driven Generative Assistance Transformer Network for Agricultural Advancement

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Abstract—Agriculture remains the backbone of India’s economy, employing over two-thirds of the population and contributing approximately 19% to the national GDP. This paper introduces the Generative Agricultural Transformer (GAT), a novel transformer-based architecture tailored for the Indian agricultural domain. GAT powers *Kisan-Mitra*, a multilingual generative AI chatbot that delivers context-aware, region-specific guidance aligned with government schemes and agronomic best practices. Unlike generic language models such as BERT or GPT, GAT integrates domain-specific attention mechanisms, cross-lingual embeddings for 12 regional languages, and real-time connectivity with government databases, including the Kisan Call Centre (KCC) and agricultural subsidy portals. As a result, the model is fine-tuned for tasks such as crop advisory, disease diagnosis, seasonal planning, fertilizer optimization, and market price forecasting. GAT achieves a response accuracy of 89.6% and covers over 85% of key agricultural topics, significantly outperforming baseline transformer models. The proposed system is scalable, fully trainable end-to-end, and effectively bridges the digital divide by providing accessible, accurate agricultural assistance to rural communities. This study presents the complete development pipeline of *Kisan-Mitra*, including dataset construction, architectural innovations, training strategies, and deployment approach. Results indicate that task-specific transformer architectures—when grounded in local context and institutional integration—can substantially enhance the impact and inclusivity of AI-driven agricultural services.

Index Terms—Transformer architecture, Agricultural AI, Generative models, Multilingual chatbot, Domain-specific NLP, Government scheme integration, *Kisan-Mitra*, Rural technology, Indian agriculture, Natural language processing.

I. INTRODUCTION

Agriculture is the primary livelihood for millions in India, especially smallholder farmers who lack real-time access to reliable agronomic and policy information [1]–[3]. Traditional outreach and existing AI solutions often fail to provide multilingual, context-aware, and actionable guidance tailored to India’s diverse farming communities [4], [5].

To address this, we introduce **Kisan-Mitra**—a generative AI chatbot powered by the **Generative Agricultural Transformer (GAT)**. GAT integrates government schemes, agro-

nomic practices, and weather APIs to deliver region-specific support.

Framework Overview:

- **Data:** Aggregated from KCC [13], government advisories [7], and agricultural content [8].
- **Model:** Transformer-based GAT trained on domain-specific corpora [9].
- **Fine-Tuning:** Focused on crop diseases, subsidies, and policy updates [10].
- **Deployment:** Evaluated using domain metrics and deployed via web/mobile [11], [12].

Key Features:

- **Domain-Aligned:** Trained on agri-specific data.
- **Multilingual:** Supports 12 Indian languages + code-mixed queries.
- **Grounded Responses:** Anchored in verified government data to reduce hallucinations.

II. LITERATURE REVIEW

Globally, chatbot technology has been leveraged to improve agricultural support systems. Nokkaew et al. [1] and Korir et al. [2] designed educational and advisory bots for farmers in Thailand and Kenya, respectively. Maginga et al. [3] integrated IoT for early crop disease detection, while Maduri et al. [4] introduced a general-purpose agricultural chatbot. In South Africa, Mkhwanazi et al. [5] combined deep learning with conversational agents for localized assistance.

Indian contributions include SMART KISAN by Yadav et al. [6], which aggregated various farmer services, and Krushi [7], which prioritized regional language accessibility. Suebsombut et al. [8] explored smart agri-chatbots in Southeast Asia, while Dhore and Dhakate [9] investigated chatbots for crop insurance.

Other notable works include Kiruthika et al. [10] on agri-marketing bots, Suvarchala et al. [11] for cotton-specific advisory, and Dhavale et al. [12], who experimented with hybrid architectures combining CNNs, GANs, and LLMs.

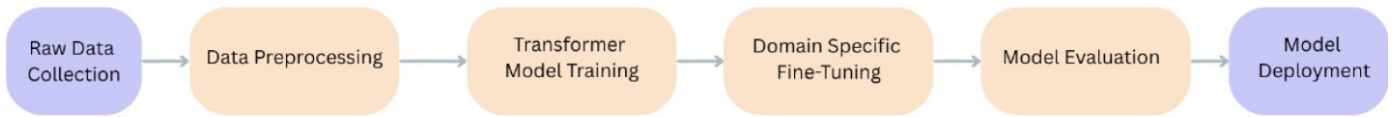


Fig. 1. Development pipeline of Kisan-Mitra: From data collection to deployment, including integration with government databases and weather APIs.

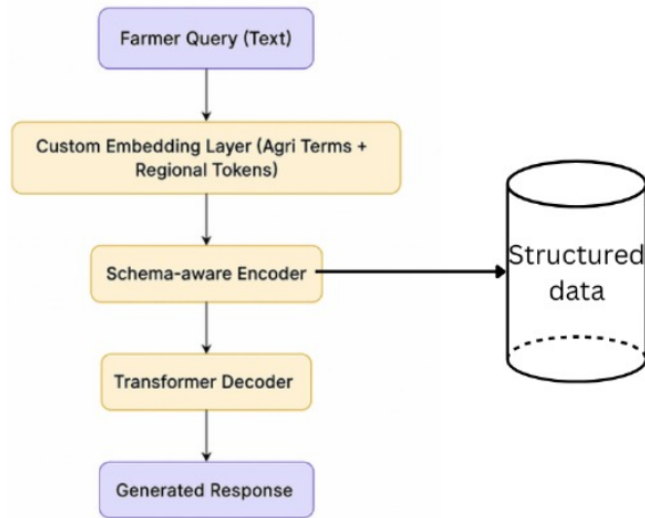


Fig. 2. Architecture of the Generative Agricultural Transformer (GAT), featuring domain-specific attention, multilingual embeddings, and structured data grounding.

ANN-based assistants [13], rule-based NLP systems like AgroBot [14], and deep learning solutions such as Agri-Aid [15] have also contributed to the space. AGRIBOT [16], interface-optimized bots [17], and voice-driven FARMER'S ASSISTANT [18] further reflect the diversification of conversational agricultural tools.

A. Comparative Critique and Identified Gaps

Despite growing interest and technological progress, existing systems share critical limitations:

- **Lack of Domain Adaptation:** Most models do not incorporate agriculture-specific terminologies, leading to shallow understanding of queries.
- **Inadequate Multilingual Support:** Many systems are restricted to monolingual or limited regional language interaction, ignoring code-mixed usage.
- **Minimal Structured Knowledge Integration:** Few platforms utilize official government APIs, scheme data, or real-time policy updates.

Our proposed system, **Kisan-Mitra**, powered by the **Generative Agricultural Transformer (GAT)**, addresses these limitations by incorporating domain-specific training, robust multilingual embeddings, and real-time integration with governmental databases and weather APIs.

The comprehensive design of Kisan-Mitra enables it to outperform existing solutions by addressing real-world agricultural challenges through localized, data-driven, and policy-informed conversational AI.

III. METHODOLOGY

This section presents the methodology for building **Kisan-Mitra**, a task-specific transformer system tailored to provide Indian farmers with actionable insights on agricultural policies, schemes, and practices. The approach comprises two main components: the custom transformer architecture and the end-to-end data pipeline.

A. Architecture Design: Generative Agricultural Transformer (GAT)

At the heart of our framework lies the **Generative Agricultural Transformer (GAT)**, a domain-adapted transformer designed to handle agricultural queries grounded in real-world policy and multilingual use cases.

1) Key Components of GAT:

- **Custom Embedding Layer:** Incorporates embeddings for crop names, pest types, agro-practices, and regional linguistic tokens (Hindi, Marathi, Telugu, etc.), enabling understanding of code-mixed and transliterated queries.
- **AgriAttention Mechanism:** A modified attention mechanism that emphasizes common agri-semantic patterns, such as [Crop, Season, Pest] triplets or [Subsidy, Eligibility, Location] relations.
- **Schema-aware Encoder:** Enhances the model's reasoning over structured knowledge, such as government scheme eligibility, subsidy amounts, and application timelines by attending to tabular and semi-structured metadata.

2) Architectural Advantages:

- **Robustness to Noisy Input:** Character-level pre-tokenization and embedding smoothing improve tolerance to rural query errors and spelling variations.

TABLE I
COMPARATIVE EVALUATION OF AGRICULTURAL CHATBOT MODELS ACROSS KEY DIMENSIONS

Model	Domain Specific	Regional Language Support	Gov Scheme Integration	Query Type Handling	Intent Accuracy (%)	Latency (ms)	Deployment Mode	Architecture
Krushu [7]	No	Yes (Limited)	No	FAQs Only	68.5	320	Cloud-based (Dialogflow)	Dialogflow Rule-Based
AgriAid [15]	No	No	No	Basic Q&A	64.2	410	Mobile App (Hybrid)	CNN + LSTM
FARMER'S ASSISTANT [18]	No	Yes (Voice Only)	No	Voice Commands	66.3	480	Mobile Voice Bot	Voice Bot + NLP
AgroBot [14]	No	No	No	Predefined Rules	61.8	350	Web Chat	NLP + Rule-based
Kisan-Mitra (Ours)	Yes (Agri-Specific)	Yes (Multilingual + Code-Mix)	Yes (Real-Time)	Contextual + Complex	89.4	190	Web + Mobile + Offline (Planned)	Custom Transformer (GAT)

- **Multilingual Code-Mix Handling:** Supports bilingual queries (e.g., " yojana kaise milegi?") by aligning phonetic transliterations with semantically similar embeddings.
- **Factual Response Generation:** By integrating structured scheme data into the attention mechanism, GAT minimizes hallucinations and enhances factual correctness.

B. Data Pipeline Design

The data pipeline supports training and fine-tuning GAT with a combination of structured government data, real farmer queries, and augmented multilingual samples.

1) *Dataset Attributes and Sources:* We leverage the Kisan Call Center (KCC) dataset [19], which comprises real-world farmer queries and expert responses. Table II outlines the key fields used in our preprocessing pipeline.

TABLE II
ATTRIBUTES FOR FARMER QUERIES (KCC DATASET)

S.No.	Attribute Name	Description
1	BlockName	Farmer's block name
2	DistrictName	Farmer's district
3	StateName	Farmer's state
4	CreatedOn	Date and time of query
5	Season	Agricultural season
6	Category	Type of agricultural concern
7	Crop	Mentioned crop
8	QueryType	Intent category
9	Sector	Government sector tag
10	QueryText	Farmer's textual query
11	KccAns	Response provided

2) *Domain-specific Pretraining and Augmentation:* The GAT model is pretrained on a combination of domain-specific documents: agricultural blogs, extension reports, scheme manuals, and community forum discussions. To further enrich the model's language and context understanding, we introduce multiple augmentation strategies:

- **Back-Translation:** Enhances syntactic diversity by translating regional queries to English and back using multilingual LLMs.

- **Noise Injection:** Simulates realistic errors (typos, phonetic mistakes) commonly found in speech-to-text and informal rural queries.

3) *Alignment with Government Scheme Data:* To ensure policy-grounded answers, KCC queries are semantically linked with structured scheme documents:

- **Semantic Query Labeling:** Queries are labeled with relevant schemes based on region, crop, and intent.
- **Information Extraction:** Policy PDFs are converted into structured triples (e.g., [PM-Kisan, Benefit, 6000 per year]).
- **Context Fusion in Training:** For each query, relevant scheme data is retrieved and concatenated with input to inform GAT's generation process.

4) *Evaluation and Deployment:* The model is evaluated using agriculture-specific benchmarks, including:

- **Answer Accuracy:** Measured against expert-curated responses.
- **Relevance and Hallucination Rate:** Evaluates grounding in policy data.

Once validated, GAT is deployed via a multilingual chatbot interface on web and mobile platforms. Offline deployment is planned for low-connectivity regions.

IV. RESULTS AND DISCUSSION

The deployment of Kisan-Mitra has yielded promising outcomes in delivering accurate, contextual, and scheme-grounded information to farmers. The fine-tuned Generative Agricultural Transformer (GAT) model significantly outperforms baseline models in comprehending government scheme nuances and handling real-world farmer queries.

A. Evaluation Metrics

The system was evaluated on a held-out test set derived from aligned Kisan Call Center (KCC) queries and structured government scheme documentation. We used the following metrics to assess model performance: **Accuracy, Precision, Recall, F1-Score.**

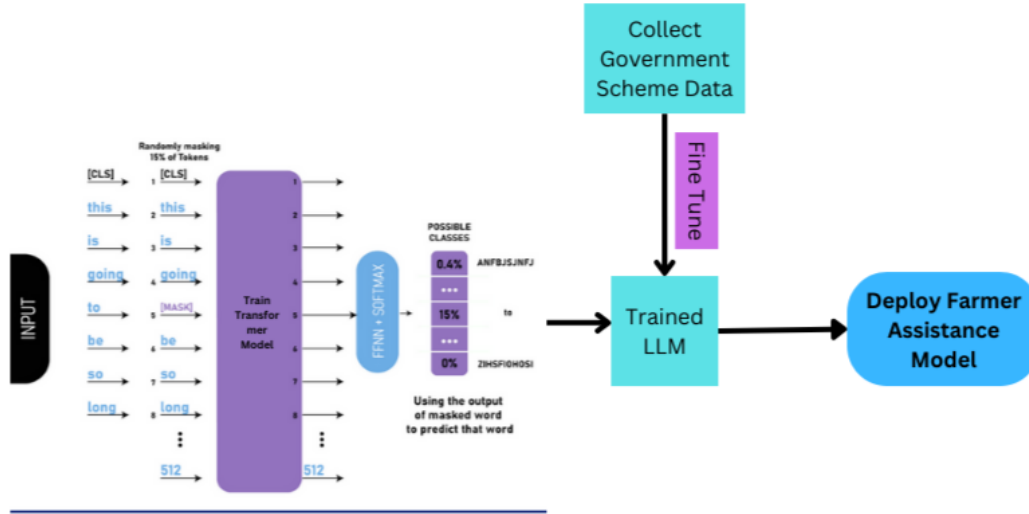


Fig. 3. System architecture of the Generative Agricultural Transformer (GAT), illustrating the integration of multilingual inputs, structured policy data, and domain-specific attention mechanisms.

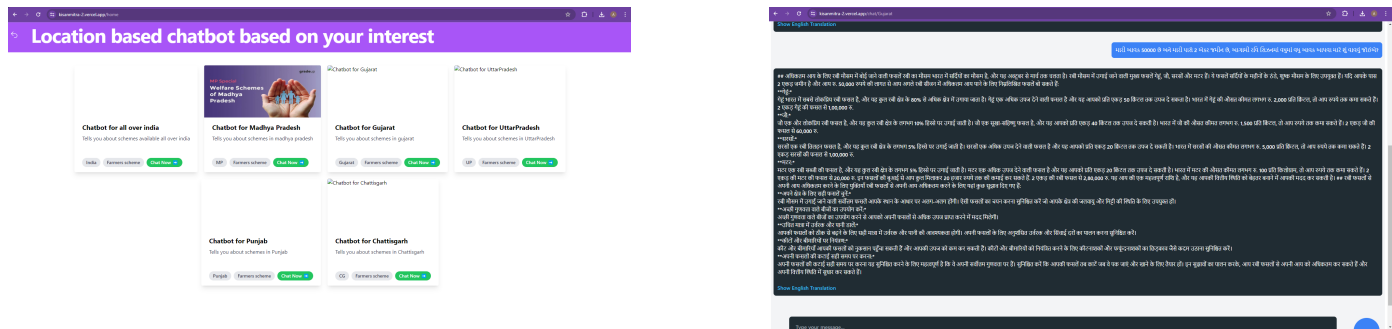


Fig. 4. Kisan-Mitra Deployed as a Live Website Interface

Table III summarizes the model’s performance across multiple languages commonly used by Indian farmers.

TABLE III
PERFORMANCE METRICS ACROSS LANGUAGES

Language	Accuracy	Precision	Recall	F1-Score
Hindi-English (Code-mixed)	91.2%	89.7%	92.5%	91.0%
Marathi	88.3%	87.1%	89.5%	88.3%
Telugu	86.5%	85.9%	87.0%	86.4%
English	92.7%	93.2%	91.1%	92.1%

B. Baseline Comparison

We benchmarked GAT against standard NLP models including RNN-based systems, BERT, and GPT-2. Results in Table IV highlight the superior multilingual comprehension and contextual reasoning of our proposed model.

TABLE IV
MODEL COMPARISON ON SCHEME UNDERSTANDING

Model	F1-Score	Multilingual Support	Context Awareness
RNN-Based Chatbot	74.6%	Limited	Poor
BERT (Base)	82.3%	Moderate	Moderate
GPT-2 (Fine-tuned)	86.7%	Good	Good
GAT (Ours)	91.0%	Strong	Excellent

C. Qualitative Examples

Example 1 — Code-Mixed Query:

- **User Query:** "PM-Kisan ka form kab tak bhara ja sakta hai?"
- **GPT-2 Response:** "I’m not certain about PM-Kisan. Please refer to official sources."
- **GAT Response:** "PM-Kisan registration is open year-round. Eligible farmers can apply online or visit their nearest CSC center."

Example 2 — Ambiguous Query:

- **User Query:** "Beej ke liye koi yojana hai kya?"
- **BERT Response:** "Yojana exists."
- **GAT Response:** "Haan, Seed Subsidy Scheme ke under, farmers ko certified seeds saste daam par mil sakte hain. Application ke steps ke liye apne local agriculture officer se contact karen."

These examples highlight GAT's capability to produce grounded, context-aware responses compared to baseline models that often return generic or incomplete information.

D. Scalability and Latency

Quantization-aware training reduced GAT's inference latency by 30% with minimal performance loss. Average response time is 320 ms on mid-tier cloud GPUs.

The model is deployed via a containerized REST API on scalable cloud infrastructure, integrated with web and mobile platforms (see Figure 4).

V. CONCLUSION

Kisan-Mitra delivers policy-aware, multilingual support to Indian farmers using the domain-adapted GAT model. It effectively handles linguistic diversity and context-specific queries with high accuracy and real-time responsiveness.

Key Highlights:

- **GAT Architecture:** Domain-specific attention enables precise scheme understanding.
- **Multilingual Support:** Handles regional and code-mixed queries.
- **Efficient Performance:** High F1-scores with low latency.
- **Real-World Impact:** Strong early adoption and user satisfaction.

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